Literature status of the Myers–Nakai problem for Banach–Finsler manifolds

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Status. Literature already answered. This is a retrieval and identification note, not a new proof or counterexample.

Source question

Jiménez-Sevilla and Sánchez-González [1], arXiv:1012.4770, state in the Introduction (PDF page 3) that an interesting open problem is whether an infinite-dimensional version of the Myers–Nakai theorem can be obtained for a class of infinite-dimensional complete Finsler manifolds. Their formulation intentionally asks only for "a certain class," rather than for all Banach–Finsler manifolds.

Exact later answer

The authors' sequel with Jaramillo [2], arXiv:1108.5403, supplies precisely such a class. Its Theorem 2.3 (PDF page 8) says that if M, N are complete, C^k-uniformly bumpable C^k Finsler manifolds modeled on k-admissible Banach spaces, then

$$M \text{ and } N \text{ are weakly } C^k \text{ Finsler equivalent} \quad \Longleftrightarrow \quad C_b^k(M) \text{ and } C_b^k(N) \text{ are isometrically algebra-isomorphic.}$$

Every such algebra isomorphism is composition with a weak C^k Finsler isometry. For $k \geq 2$, that map is a C^{k-1} Finsler isometry.

This is genuinely infinite dimensional. Corollary 2.7 (PDF page 11) covers complete C^1-uniformly bumpable Finsler manifolds modeled on weakly compactly generated Banach spaces. Corollary 2.8, on the same page, covers complete separable C^k Finsler manifolds modeled on Banach spaces possessing a Lipschitz C^k-smooth bump function. Thus the later paper directly realizes the source paper's requested infinite-dimensional Myers–Nakai program.

Scope and the other detector signals

The sequel does not assert a theorem for every Banach–Finsler manifold. At $k = 1$ it obtains weak C^1 equivalence because a fully general Finsler Myers–Steenrod regularity theorem is unavailable; this limitation is stated immediately after the proof of Theorem 2.3.

The two "we do not know" passages in arXiv:1012.4770 concern a different renorming-uniformity issue: whether the approximation constant in property $(*^1)$ can be chosen uniformly over all equivalent norms, and whether the smooth Lipschitz extension theorem remains true without that hypothesis. Theorem 2.3 does not resolve those questions. They should not be conflated with the advertised Myers–Nakai open problem cleared here.

References

[1] M. Jiménez-Sevilla and L. Sánchez-González, *On some problems on smooth approximation and smooth extension of Lipschitz functions on Banach–Finsler manifolds*, arXiv:1012.4770 (2010), Nonlinear Analysis 74 (2011), 3487–3500.

[2] J. A. Jaramillo, M. Jiménez-Sevilla, and L. Sánchez-González, *Characterization of a Banach–Finsler manifold in terms of the algebras of smooth functions*, arXiv:1108.5403 (2011).